

HW 2: due Monday January 30

Part I (Reading and Resources):

1. (3 pts) Read Scheinerman **3rd ed.**, sections 5–7. For each section, describe one thing you learned and/or enjoyed reading. Also read the description of solving syllogisms found at <http://www.math.hawaii.edu/~hile/math100/logice.htm> (no reading response needed for the webpage).
2. (1 pt) List all resources you end up using to help you solve the homework problems below (e.g., people, TAs, websites)

Part II:

1. Read the handout “Writing Mathematics Well” in Resources. For each of the six bullet point guidelines at the top, rate how much your overall Lion and the Unicorn solution from Homework 1 followed that guideline. (I.e. your answer is six ratings total; don’t rate the individual parts of the problem.). Use any rating scale you like, e.g. 1 to 5, or “Oops” “Partly” “Nailed it”, or smiley/frowny faces.

2. Suppose right now it is raining, and I am drinking tea and reading a book, somewhere which is not my bedroom.

Based on the above information, classify each of the following as being currently True, False, or Cannot be Determined. Explain your answers (a sentence for each is sufficient.) Note you are only analyzing the **current, fixed** truth values; you do not have to think about what the truth values might be at other times, or in general.

- (a) If I am drinking tea right now, then I am reading a book.
 - (b) If I am drinking tea right now, then I am in my bedroom.
 - (c) If it is not raining right now, then I am drinking tea.
 - (d) If it is not raining right now, then I am riding a bicycle.
 - (e) If I am wearing purple right now, then I am drinking tea.
 - (f) If I am reading a book right now, then I am at the library.
3. Solve the following syllogisms by first rewriting each statement as a logical implication, then combining logical implications by transitivity, and finally writing the overall conclusion in English.
 - (a)
 - The only articles of food that my doctor allows me are not very rich.
 - Nothing that agrees with me is unsuitable for supper.
 - Wedding cake is always very rich.

- My doctor allows me all articles of food that are suitable for supper.

Universe = “articles of food”; D = “doctor allows”, R = “is very rich”, A = “agrees with me”, S = “is suitable for supper”, W = “is wedding cake.”

- (b)
- Nobody, who really appreciates Beethoven, fails to keep silence while the Moonlight-Sonata is being played;
 - Guinea-pigs are hopelessly ignorant of music;
 - No-one, who is hopelessly ignorant of music, ever keeps silence while the Moonlight-Sonata is being played.

Universe = “creatures”; *G* = guinea-pigs; *I* = hopelessly ignorant of music; *S* = keeping silence while the Moonlight-Sonata is being played; *B* = really appreciating Beethoven.

Part III, from *Scheinerman* do the following problems. “2nd{x.y}” refers to problem x.y in the 2nd edition and “3rd{x.y}” refers to problem x.y in the 3rd edition.

Please note that what we are most interested in is the clarity of your explanations, not just the correct answer.

3rd{5.6, 5.9, 5.15} OR 2nd{4.5, 4.6, 4.11}

AND

3rd{6.1, 6.2, 6.11} OR 2nd{5.1, 5.2, 5.7}

AND

3rd{7.3, 7.9, 7.10c, 7.11b, 7.15, Bonus: 7.17, 7.18} OR
2nd{6.3, 6.9, 6.10c, 6.11b, 6.14, Bonus: 6.16, 6.17}

Final question (1 pt): How much time did you spend on this homework, in hours?

BONUS! Office Hour Scavenger Hunt. Visit different TAs’ office hours this week and collect Scavenger Hunt problems to solve. Each TA has a different problem. Each correct answer gets you 1 bonus pt (max of 4 pts credit, but you can try as many as you like.) Please write the TA’s name for each answer you submit.